

7.6 Nitrogen Compounds (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.6 Nitrogen Compounds (A Level Only)
Difficulty	Medium

Time allowed: 80

Score: /62

Percentage: /100

Question 1a

Three nitrogen containing molecules, ammonia, NH_3 , phenylamine, $\text{C}_6\text{H}_5\text{NH}_2$ and *N*-methylethylamine, $\text{CH}_3\text{CH}_2\text{NHCH}_3$, are drawn below respectively in Fig. 1.1.

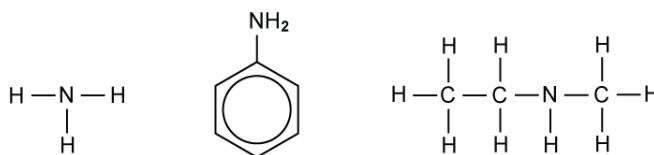


Fig. 1.1

i)

List the three amine molecules drawn in Fig. 1.1 in order of increasing base strength.

[1]

ii)

Explain your answer to part (b) (i).

[6]

[7 marks]

Question 1b

Phenylamine, $\text{C}_6\text{H}_5\text{NH}_2$ shown in Fig. 1.1 in part (a) can be produced from nitrobenzene, $\text{C}_6\text{H}_5\text{NO}_2$.

Name the type of reaction and suggest suitable reagents and conditions for this conversion.

[3 marks]

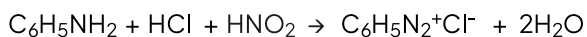
Question 1c

Azo dyes are organic compounds, which are largely used in the treatment of textiles and leather products, as well as in food. Phenylamine, $\text{C}_6\text{H}_5\text{NH}_2$, can be used in the manufacture of azo dyes.

The manufacturing process is outlined below.

Step 1

Phenylamine is dissolved in HCl to produce a diazonium salt. A diazonium salt contains two nitrogen atoms joined together by a triple bond. The reaction for this process is as follows:

**Step 2**

This solution is then slowly added to an alkaline solution of a phenol coupling agent to form the dye.

i)

The diazonium salt, $\text{C}_6\text{H}_5\text{N}_2^+\text{Cl}^-$, is an unstable compound. Suggest a condition that could be added to ensure that the salt would not break down during the reaction.

[1]

ii)

Draw the structure of the diazonium salt formed in **Step 1**, showing the displayed formula of the nitrogen containing group.

[1]

[2 marks]

Question 1d

Phenol, C_6H_5OH , is an aromatic organic compound which is also crucial in the manufacture of azo dyes.

The final step of the azo dye production involves pairing up the diazonium salt with a phenol compound as a coupling agent.

Suggest a structure for the azo dye if the coupling agent used is 2,6-dimethylphenol.

[2 marks]

Question 2a

Amines will react with acid chlorides to produce a different nitrogen containing product.

i)
Name and outline the mechanism between ethanoyl chloride, CH_3COCl and methylamine, CH_3NH_2 .

[5]

ii)
Give the IUPAC name of the compound formed in the reaction.

[1]

[6 marks]

Question 2b

Methylamine can be prepared from chloromethane.

i)
Give the other reagent, and any reaction conditions necessary for the preparation of methylamine.

[3]

ii)
Give the name of the mechanism between chloromethane and the other reagent to form methylamine.

[1]

[4 marks]**Question 2c**

Chloroethane can also react with methylamine.

i)
Draw the displayed formula of the organic compound formed in this reaction.

[1]

ii)
Give the classification of this compound.

[1]

[2 marks]

Question 2d

Two separate solutions of methylamine, CH_3NH_2 and dimethylamine, $(\text{CH}_3)_2\text{NH}$, were left unlabelled in a laboratory.

i)

Suggest how, other than smell, they could be distinguished from one another.

[2]

ii)

Give an explanation to your answer to part (i).

[2]

[4 marks]

Question 3a

Much research has been carried out in recent years investigating the exact structure of silk. The silk of a spider's web is at least five times as strong as steel, and twice as elastic as nylon. A silk fibre is composed of many identical protein chains, which are mainly made from the amino acids glycine, alanine and serine, shown in Fig. 3.1, with smaller amounts of four other amino acids.

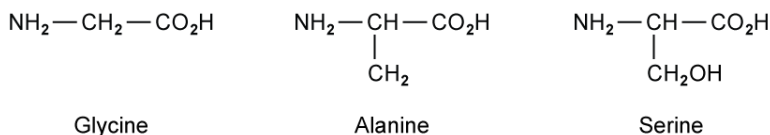


Fig. 3.1

Amino acids can exist as zwitterions. Draw the zwitterion structure for glycine.

[1 mark]

Question 3b

Amino acids can act as acids or bases. Write equations to show:

- i)
the reaction between alanine and HCl (aq),

[1]

- ii)
the reaction between serine and NaOH (aq).

[1]

[2 marks]**Question 3c**

- i)
Draw the structural formula of a portion of the silk protein, showing three main acid residues. Label a peptide bond on your structure.

[3]

- ii)
State the polymer **type** of silk protein.

[1]

[4 marks]

Question 3d

The M_r of a silk protein molecule is about 600,000. Assuming it is made from equal amounts of the three amino acids in Fig. 3.1, calculate the average number of amino acid residues in the protein chain.
Show your working.

[$M_r(\text{glycine}) = 75$; $M_r(\text{alanine}) = 89$; $M_r(\text{serine}) = 105$]

number of residues

[3 marks]

Question 4a

Phenylamine can be converted into the organic compounds **A** and **B** as shown in Fig. 4.1.

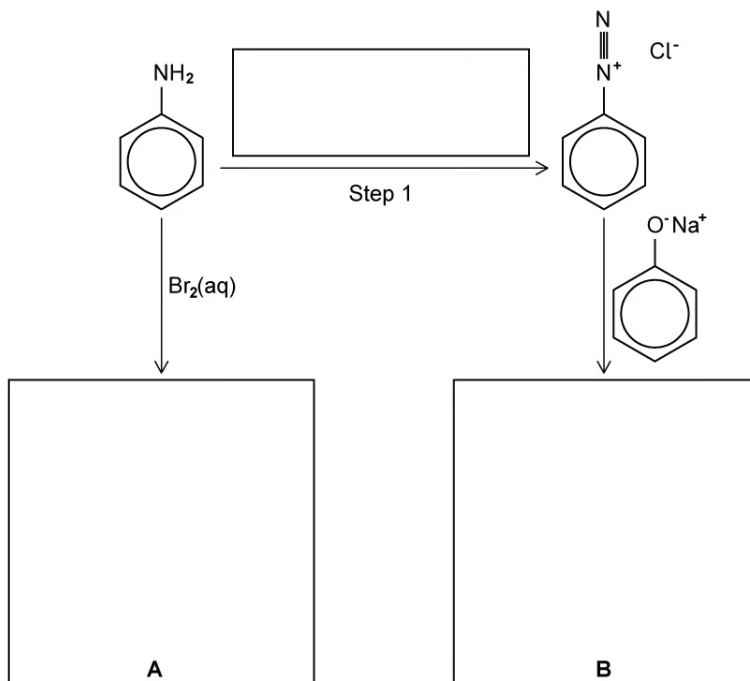


Fig. 4.1

- i)
Suggest the structural formulae of **A** and **B** in the boxes in Fig. 4.1.
- ii)
Suggest suitable reagents and conditions for step 1, and write them in the box over the arrow in Fig. 4.1.

[2]

[2]

[4 marks]

Question 4b

When phenylamine is treated with propanoyl chloride a white crystalline compound, **C**, $\text{C}_9\text{H}_{11}\text{NO}$, is formed.

i)

Name the functional group formed in this reaction.

[1]

ii)

Calculate the percentage by mass of nitrogen in **C**.

[1]

percentage = %

iii)

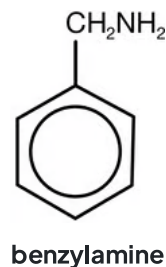
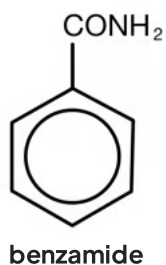
Draw the structural formula of **C**.

[1]

[3 marks]

Question 4c

Benzamide can be reduced to form benzylamine.



- i)
State which is the weaker base, benzamide or benzylamine. Explain your answer.

[3]

- ii)
Give a suitable reducing agent required for this reduction.

[1]

[4 marks]

Question 4d

When benzamide is refluxed with acid, hydrolysis occurs.

i)

Draw the structural formula of the resulting organic compound.

[1]

ii)

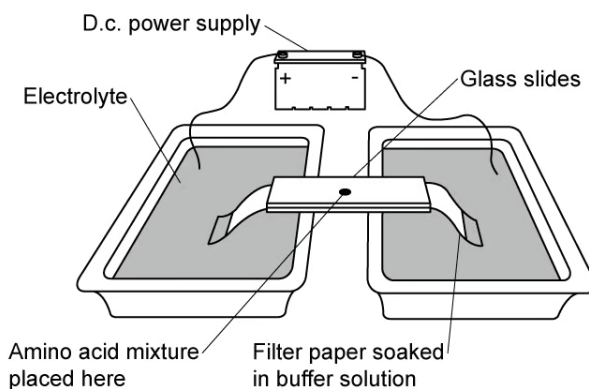
State the name of the other product formed during this hydrolysis reaction.

[1]

[2 marks]

Question 5a

A mixture of amino acids may be separated using electrophoresis. A typical practical set-up is shown in the diagram.



When the power supply is switched on, some amino acids may not move, but remain stationary. Suggest an explanation for this observation.

[2 marks]

Question 5b

The amino acid glycine has the formula $\text{H}_2\text{NCH}_2\text{CO}_2\text{H}$. Identify the species formed on the filter paper if glycine moves to the left (positive) end of the filter paper.

[1 mark]

Question 5c

The result shown in Fig. 5.1 was obtained from another electrophoresis.

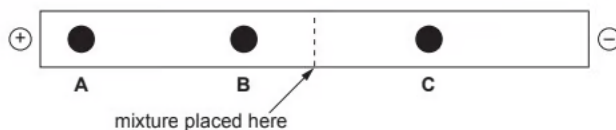


Fig. 5.1

What can be deduced about the relative sizes of, and charges on, the amino acid species **A**, **B** and **C**?

amino acid	relative size	charge
A		
B		
C		

[3 marks]

Question 5d

A tripeptide was formed from three different amino acids, alanine, serine and valine. The structural formulae of the amino acids are shown in Table 5.1.

Table 5.1

amino acid	structural formulae
alanine	$\text{H}_2\text{NCH}(\text{CH}_3)\text{CO}_2\text{H}$
serine	$\text{H}_2\text{NCH}(\text{CH}_2\text{OH})\text{CO}_2\text{H}$
valine	$\text{H}_2\text{NCH}(\text{CH}(\text{CH}_3)_2)\text{CO}_2\text{H}$

i)
How many different tripeptides can be made using **one** molecule of **each** of the amino acids in Table 5.1?

[2]

ii)
Draw the tripeptide, showing the peptide bonds in displayed form.

[1]

[3 marks]

Model Answers: Medium

1a

a)

i) The three molecules in order of increasing base strength is:

- Phenylamine < ammonia < *N*-methylethylamine

OR

Phenylamine is the weakest **AND** *N*-methylethylamine is the strongest base; [1 mark]

ii) Explaining the reason why:

- Base strength depends on the availability of the lone pair on the N atom; [1 mark]
- N*-methylethylamine is strongest due to the positive inductive effect from alkyl groups

OR

Alkyl groups on *N*-methylethylamine push electrons / negative charge towards the N; [1 mark]

- The lone pair is more / most available to accept a proton in *N*-methylethylamine; [1 mark]
- In phenylamine, the N atom is bonded directly to the benzene ring; [1 mark]
- The lone pair on the N becomes delocalised within / into the benzene ring

OR

Overlap of the lone pair on N and the benzene ring electrons; [1 mark]

- The lone pair on N is no longer / not fully available to accept a proton; [1 mark]

[Total: 7 marks]

- The greater the electron density on the nitrogen atom in an amine the more likely the lone pair is to be available for bonding
 - Alkyl groups have a tendency to push electrons away from themselves and onto the nitrogen atom
 - There are two alkyl groups in *N*-methylethylamine, a secondary amine, and

none in ammonia

- This is known as the positive inductive effect
 - The more alkyl groups there are the more available the lone pair of electrons is on the nitrogen atom
- In phenylamine, the lone pair on the nitrogen atom is drawn towards the delocalisation in the ring and this interaction makes the lone pair less available to bond with an incoming proton (H^+ ion)

1b

a)

i) The name of the type of reaction and suitable reagents and conditions for the conversion of nitrobenzene to phenylamine are:

- Reduction; [1 mark]

Method 1

- Sn / tin **AND** concentrated HCl; [1 mark]
- Heat **AND** NaOH solution; [1 mark]

Method 2

- H_2 ; [1 mark]
- Ni / Pt / Pd catalyst; [1 mark]

Method 3

- LiAlH_4 ; [1 mark]
- Dry ether; [1 mark]

[Total: 3 marks]

- Once the benzene ring has undergone nitration, the NO_2 group can then be reacted with appropriate reducing agents
- Nitration of a benzene ring followed by the reduction of the NH_2 group is a common pathway so it is worth making sure you have learned this thoroughly
- Method 1 is the method that you need to know, but the other methods are all suitable methods so are creditworthy
- LiAlH_4 in dry ether is an almost universal reducing agent and works in most

examples of organic reduction. It can be used to convert nitrile groups ($-\text{CN}$) to amino groups ($-\text{NH}_2$) so is very important in the production of amines

1c

c)

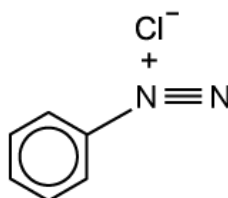
i) A condition that could be added to ensure that the salt would not break down during the reaction is:

- Completed at a low temperature

OR

Done in ice / ice cold conditions; [1 mark]

ii) The structure of the organic compound formed in the ice-cold acidic mixture is:



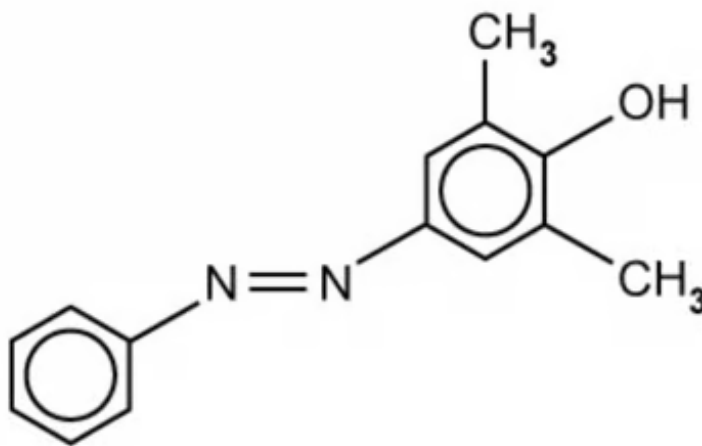
- Structure showing nitrogen containing group with triple bond **AND** positive charge on correct N atom **AND** negative charge on the Cl; [1 mark]

[Total: 2 marks]

- In order for the nitrous acid to be used in this process it must be made in situ as it is an unstable compound
 - The reaction must also be kept at a low temperature to prevent this from decomposing
- The equation for the reaction is:
 - $\text{NaNO}_2(\text{aq}) + \text{HCl}(\text{aq}) \rightarrow \text{HNO}_2(\text{aq}) + \text{NaCl}(\text{aq})$
- The acid can then react with the amine to produce a diazonium salt
- A nitrogen atom that has four bonds will have a positive charge as it is electron deficient

1d

d) A structure for the azo dye if the coupling agent used is 2,6-dimethylphenol is:



- Correct bond link between N atom and the coupling agent; [1 mark]
- Correct overall structure; [1 mark]

[Total: 2 marks]

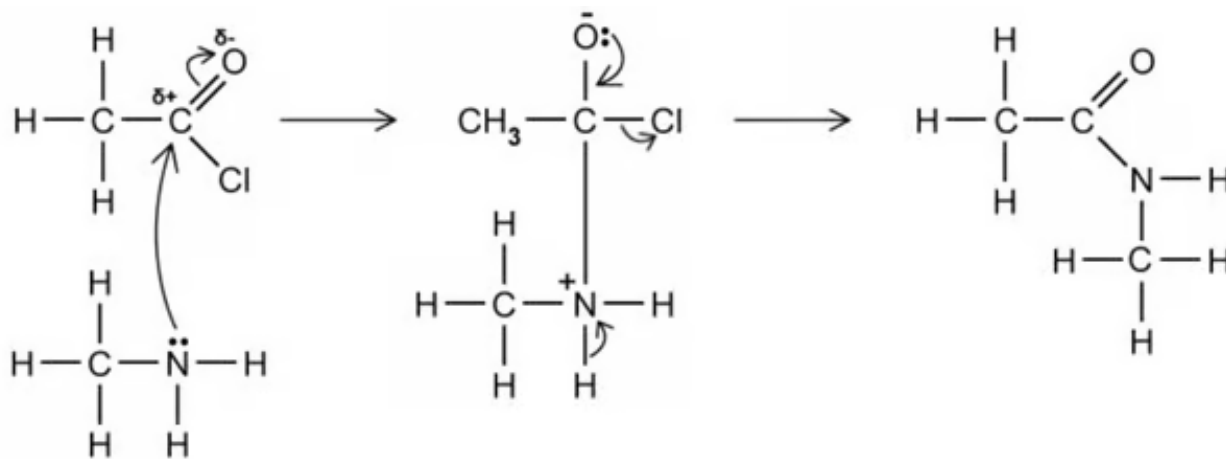
- The -OH group should be the opposite side of the nitrogen containing group
 - A coupling agent is an arene compound that will bond to the diazonium salt
 - The diazonium salt acts as an electrophile
 - Coupling of diazonium salts with different aromatic compounds forms dyes of different colours

2a

a)

i) The name and outline of the mechanism between ethanoyl chloride and methylamine is:

- (Nucleophilic) addition elimination; [1 mark]



- Curly arrow from lone pair on N atom to delta positive C atom **AND** curly arrow from $\text{C}=\text{O}$ bond to O atom; [1 mark]
- Correct intermediate structure; [1 mark]
- Curly arrow from lone pair on O atom to $\text{C}-\text{O}$ bond **AND** curly arrow from the $\text{C}-\text{Cl}$ bond to the Cl atom; [1 mark]
- Curly arrow from $\text{N}-\text{H}$ bond to positively charged N atom; [1 mark]
- Correct product structure; [1 mark]

ii) The IUPAC name of the compound formed in this reaction is:

- *N*-methylethanamide; [1 mark]

[Total: 6 marks]

- An *N*-substituted amide is produced during the vigorous reaction between an amine and an acyl chloride
- **Always** show the lone pair and negative charge on the oxygen atom in the second step
- The nitrogen in the intermediate must have a positive charge as it has four bonds
- The final stage of the mechanism involves an H atom being lost. This is not required in the mechanism, though it can be removed by the $:\text{Cl}^-$ ion or by another propylamine molecule
- The product is an *N*-substituted amide
 - The '*N*' shows that there is substitution on the nitrogen atom and nowhere else
- This is an example of a condensation reaction as two molecules join together and in the process eliminate the small molecule, HCl

2b

b)

i) The other reagent, and any reaction conditions necessary for the preparation of methylamine are:

- Ammonia / NH_3 ; [1 mark]
- Excess hot ethanolic ammonia; [1 mark]
- Under pressure; [1 mark]

ii) The name of the mechanism between chloromethane and the other reagent to form methylamine is:

- Nucleophilic substitution; [1 mark]

[Total: 4 marks]

- The conditions used prevent the formation of secondary and tertiary amines as the excess ammonia reduces the likelihood that methylamine will attack

- Curly arrow from N-H bond to positively charged N atom; [1 mark]
- Correct product structure; [1 mark]

ii) The IUPAC name of the compound formed in this reaction is:

- *N*-methylethanamide; [1 mark]

[Total: 6 marks]

- An *N*-substituted amide is produced during the vigorous reaction between an amine and an acyl chloride
- **Always** show the lone pair and negative charge on the oxygen atom in the second step
- The nitrogen in the intermediate must have a positive charge as it has four bonds
- The final stage of the mechanism involves an H atom being lost. This is not required in the mechanism, though it can be removed by the :Cl⁻ ion or by another propylamine molecule
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2b

b)

i) The other reagent, and any reaction conditions necessary for the preparation of methylamine are:

- Ammonia / NH₃; [1 mark]
- Excess hot ethanolic ammonia; [1 mark]
- Under pressure; [1 mark]

ii) The name of the mechanism between chloromethane and the other reagent to form methylamine is:

- Nucleophilic substitution; [1 mark]

[Total: 4 marks]

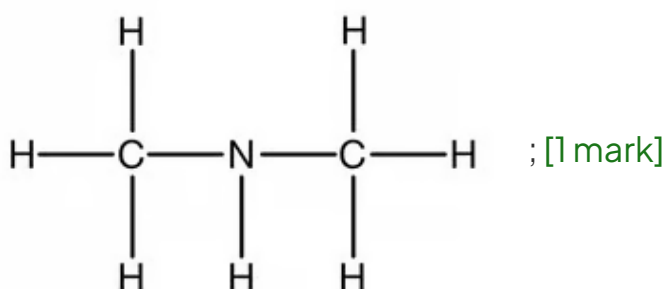
- The conditions used prevent the formation of secondary and tertiary amines as the excess ammonia reduces the likelihood that methylamine will attack chloromethane, as methylamine can also act as a nucleophile
- The excess ammonia also reacts with the HCl produced in the reaction, forming NH_4Cl
 - This removes HCl from the reaction mixture preventing it from reacting with methylamine
- This is a nucleophilic substitution reaction in which the nitrogen lone pair in ammonia acts as a nucleophile and replaces Cl in the chloromethane

2c

c)

i) The displayed formula of the organic compound formed in this reaction is:

•



ii) The classification of this compound is:

- Secondary **AND** amine; [1 mark]

[Total: 2 marks]

- When a halogenoalkane reacts with a primary amine, a secondary amine is formed
- The halogen atom is replaced by the nitrogen in the primary amine
- This is another example of nucleophilic substitution
- The product formed is a secondary amine as it has a -NH group attached to 2 alkyl groups
 - The product is called dimethylamine but you are not required to name secondary amines

2d

d)

i) Methylamine and dimethylamine could be distinguished from one another by:

- Use a pH meter; [1 mark]
- Dimethylamine will give a higher pH reading than methylamine; [1 mark]

ii) Explaining the answer to part (i):

- Base depends on the availability of the lone pair on the nitrogen atom; [1 mark]
- Dimethylamine is the strongest base

AND

Due to positive inductive effect / electron donating nature of two alkyl groups bonded to the N atom; [1 mark]

[Total: 4 marks]

- Secondary amines, such as dimethylamine, are stronger bases than primary amines, such as methylamine
 - This is due to the fact that there are more alkyl groups bonded to the nitrogen atom pushing electron density towards it
 - This in turn would make the lone pair of electrons on the nitrogen more available for accepting a proton
 - This is called the positive inductive effect
- In terms of base strength tertiary amines > secondary amines > primary amines > ammonia

3a

a) The zwitterion structure for glycine is:

- $\text{NH}_3^+\text{CH}_2\text{CO}_2^-$; [1 mark]

[Total: 1 mark]

- Each amino acid can interact within itself where the hydrogen ion from the CO_2H is transferred to the -NH_2 group
- This results in the formation of an ion with both a positive and negative charge,

but which is overall uncharged – this is a zwitterion

- To draw the structural formula of glycine, move the H from the $\text{-CO}_2\text{H}$ to the -NH_2 group and add a positive charge to the resulting NH_3 and a negative charge to the -CO_2
- **Remember:** Do not show the single bonds in the structural formula

3b

b)

i) The equation for the reaction between alanine and HCl (aq) is:

- $\text{NH}_2\text{CH}(\text{CH}_3)\text{CO}_2\text{H} + \text{HCl} \rightarrow \text{ClNH}_3\text{CH}(\text{CH}_3)\text{CO}_2\text{H}$; [1 mark]

ii) The equation for the reaction between serine and NaOH (aq) is:

- $\text{NH}_2\text{CH}(\text{CH}_2\text{OH})\text{CO}_2\text{H} + \text{NaOH} \rightarrow \text{NH}_2\text{CH}(\text{CH}_2\text{OH})\text{CO}_2\text{Na} + \text{H}_2\text{O}$; [1 mark]

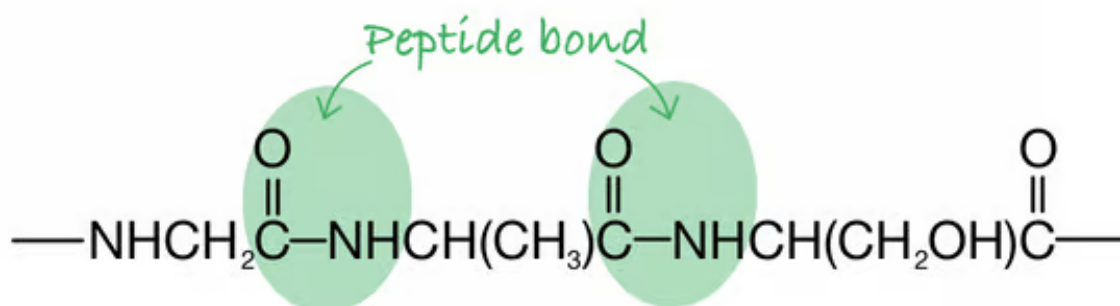
[Total: 2 marks]

- The amino acids will exist as zwitterions
- When HCl is added to alanine, the H^+ ion will transfer to the -CO_2^- on the zwitterion and the Cl^- ion will transfer to the NH_3^+ on the zwitterion
- When NaOH is added to serine, an H^+ from the -NH_3^+ on the zwitterion ion will transfer to the OH^- ion, forming water, and the Na^+ will transfer to the -CO_2^- on the zwitterion

3c

c)

i) The structural formula of a portion of the silk protein, with a peptide bond shown is:



- Correct CO-NH bonding, including at least one C=O bond; [1 mark]
- At least one peptide bond labelled; [1 mark]
- 3 amino acid residues; [1 mark]

ii) The type of polymer is:

- Condensation
OR
Polyamide; [1 mark]

[Total: 4 marks]

- The amine group of one amino acid will react with the carboxylic acid group of another amino acid
- A condensation reaction occurs and a water molecule is lost
- A peptide bond, CO-NH, forms between the two amino acids
- The structural formula that you draw needs to contain 3 residues (i.e. an amino acid that has lost H₂O)
- The three residues don't all need to be different but must be either glycine, alanine or serine

3d

d) To calculate the average number of amino acid residues in the protein chain:

- Deduct 18 from each M_r value to give M_r value of 3-residues = 215; [1 mark]
- $\frac{600\,000}{215} = 2791$; [1 mark]
- $2791 \times 3 = 8372$; [1 mark]

[Total: 3 marks]

- When forming the protein chain, each amino acid will lose a water molecule, so each M_r will reduce by $(2 \times 1) + 16 = 18$
 - M_r of glycine residue in protein chain = $75 - 18 = 57$
 - M_r of alanine residue in protein chain = $89 - 18 = 71$
 - M_r of serine residue in protein chain = $105 - 18 = 87$
 - M_r of all three amino acid residues in protein chain = $57 + 71 + 87 = 215$
- The second mark is given for dividing 600 000 by the M_r value of the three residues

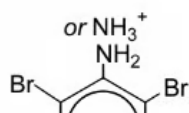
- If you incorrectly calculated the M_r to be 269 by not subtracting 18 from each M_r , then you could still gain this mark (this would give you a value of 2230)
- This value tells you how many sets of the three amino acid residues must be present in the protein chain
- As the question asks how many amino acid residues there are, you need to multiply this number by 3
 - If you didn't deduct 18 from each M_r value, then this would give 6691 as your answer - you would gain 2 marks for this

4a

a)

i) The structural formula of **A** and **B** are:

•



[1 mark]

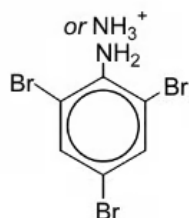
- M_r of glycine residue in protein chain = $75 - 18 = 57$
- M_r of alanine residue in protein chain = $89 - 18 = 71$
- M_r of serine residue in protein chain = $105 - 18 = 87$
- M_r of all three amino acid residues in protein chain = $57 + 71 + 87 = 215$
- The second mark is given for dividing 600 000 by the M_r value of the three residues
 - If you incorrectly calculated the M_r to be 269 by not subtracting 18 from each M_r , then you could still gain this mark (this would give you a value of 2230)
- This value tells you how many sets of the three amino acid residues must be present in the protein chain
- As the question asks how many amino acid residues there are, you need to multiply this number by 3
 - If you didn't deduct 18 from each M_r value, then this would give 6691 as your answer - you would gain 2 marks for this

4a

a)

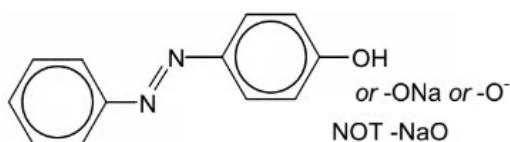
i) The structural formula of **A** and **B** are:

•



; [1 mark]

•



; [1 mark]

ii) Suitable reagents and conditions for step 1 are:

- $\text{NaNO}_2 + \text{HCl} / \text{H}_2\text{SO}_4 / \text{H}^+$
OR
 HNO_2 ; [1 mark]
- $T \leq 10^\circ\text{C}$; [1 mark]

[Total: 4 marks]

- The reaction of phenylamine with bromine occurs more readily than the reaction of benzene with bromine, which requires a halogen carrier catalyst
- The -NH_2 group in phenylamine activates the ring and the aromatic ring will react more readily with electrophiles
- The -NH_2 group is 2- and 4- directing, so the incoming electrophile will be substituted into positions 2, 4 and 6
- The resulting compound is 2,4,6-trinitrophenylamine
- When phenylamine reacts with nitric(III) acid, HNO_2 , at temperatures below 10°C , a diazonium salt is produced
 - Nitric(III) acid is unstable so needs to be freshly prepared from sodium nitrite, NaNO_2 , and a dilute acid, such as HCl
- The diazonium ion acts as an electrophile and will substitute into the benzene ring of $\text{C}_6\text{H}_5\text{O}^-$ in the 4th position
- The delocalised electrons in the π bonding systems of the two benzene rings are extended through the -N=N- which acts as a bridge between the two rings
- -N=N- is the azo group

4b

b)

i) The functional group formed in this reaction is:

- Amide; [1 mark]

ii) The percentage by mass of nitrogen in **C** is:

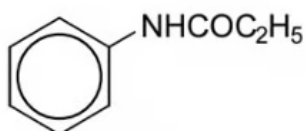
- $M_r = (9 \times 12) + (11 \times 1) + 14 + 16 = 149$

AND

$$\%N = \frac{14}{149} \times 100 = 9.4(\%); [1 \text{ mark}]$$

ii) The structural formula of **C** is:

•



; [1 mark]

[Total: 3 marks]

- Phenylamine is a primary amine
 - When a primary amine is reacted with an acyl chloride, a substituted amide is produced
- Amides have the -CONR_2 functional group
- The nitrogen in phenylamine has a lone pair of electrons that act as a nucleophile, attacking the δ^+ carbonyl carbon in the acyl chloride
 - The two molecules join together with the loss of a molecule of HCl
 - This is an addition-elimination reaction, also known as a condensation reaction
 - The resulting compound is phenyl propanamide
- To draw the structural formula, join the two molecules together by bonding the nitrogen of the phenylamine to the carbonyl carbon of the acyl chloride, removing the Cl from propanoyl chloride and removing an H attached to the N in phenylamine
- In part (ii), a correct answer scores the mark without working

4c

c) The weaker base is:

- Benzamide; [1 mark]
- (Because) the electron-withdrawing oxygen atom; [1 mark]
- Means that the lone pair on the nitrogen atom is less readily available / not available to donate to an electron-deficient species; [1 mark]

ii) A suitable reducing agent for this reduction is:

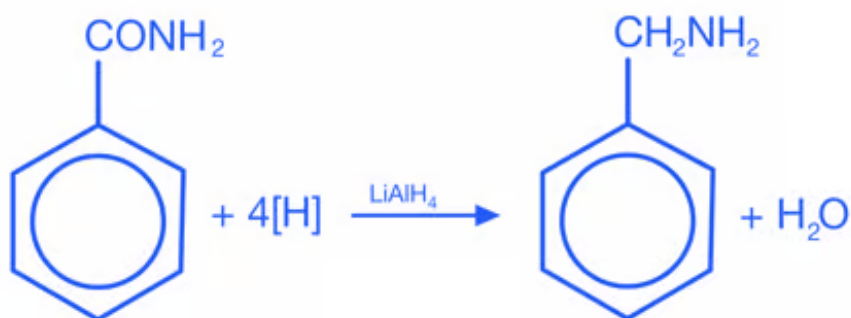
- LiAlH_4 / lithium aluminum hydride; [1 mark]

[Total: 4 marks]

- Benzylamide is a weaker base due to the electron-withdrawing oxygen atom
- This draws the electron density from the nitrogen atom which causes the lone pair to be much less readily available
- Both benzylamine and benzamide also contain the electron-withdrawing benzene ring which also draws electrons from the nitrogen atom
 - This is why benzylamine is a weaker base than, for example, propylamine
- **Careful:** Whilst you can use other reducing agents, such as NaBH_4 , for the reduction

of some organic compounds, such as carbonyl compounds, a very strong reducing agent is required to reduce the C=O group in amides

- When benzamide is reduced, benzylamine and water are produced:

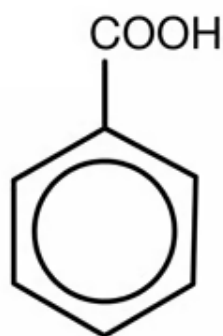


4d

d)

i) The structural formula of the resulting organic compound is:

•



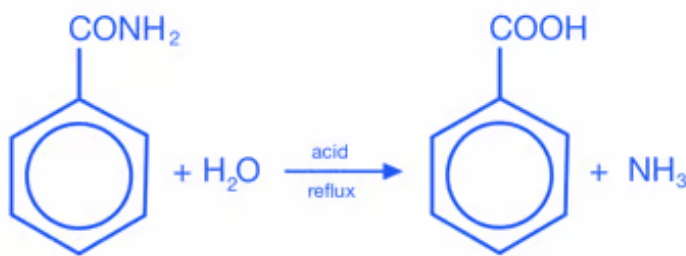
; [1 mark]

ii) The name of the other product formed during this hydrolysis reaction is:

- Ammonia; [1 mark]

[Total: 2 marks]

- When non-substituted amides are refluxed with acid or alkali, a carboxylic acid and ammonia are produced:
 - benzamide → benzoic acid + ammonia



5a

a) When the power supply is switched on, some amino acids may not move, but remain stationary because:

Any **two** from the following:

- The amino acid is uncharged / neutral / a zwitterion or charges balance / are equal; [1 mark]
- It is equally attracted by the anode / + and the cathode / -

OR

It is attracted by neither the anode or the cathode; [1 mark]

5a

a) When the power supply is switched on, some amino acids may not move, but remain stationary because:

Any **two** from the following:

- The amino acid is uncharged / neutral / a zwitterion or charges balance / are equal; [1 mark]
 - It is equally attracted by the anode / + and the cathode / -
- OR**
- It is attracted by neither the anode or the cathode; [1 mark]
- The pH of the buffer is at the isoelectric point / IEP of the amino acid; [1 mark]

[Total: 2 marks]

- When a sample of amino acids is placed between two oppositely charged electrodes
 - The positively charged amino acids will move toward the negative electrode
 - The negatively charged amino acids will move toward the positive electrode
 - Amino acids with an overall neutral charge will remain stationary
- If the pH of the buffer is at the isoelectric point of the amino acid, then neither the positively or negatively charged ions of the amino acid will dominate and the amino acid will exist as a neutral zwitterion

5b

b) The species formed on the filter paper if glycine moves to the left (positive) end of the filter paper is:

- $\text{H}_2\text{NCH}_2\text{CO}_2^- / \text{H}_2\text{NCH}_2\text{COO}^-$; [1 mark]

[Total: 1 mark]

- If a species moves to the positive side, it must be negative
- Glycine forms a negative ion by losing the H from the -COOH group

5c

c) The following can be deduced about the relative sizes of, and charges on, the

amino acid species **A**, **B** and **C**:

amino acid	relative size	charge
A	small(est) / 1	-
B	large(est) / 3	-
C	middle / 2	+

- Each correct row; [1 mark]

[Total: 3 marks]

- The direction of travel depends on the charge on the ion
 - If the amino acid species moves toward the positive side, it is negative
 - If the amino acid species moves toward the negative side, it is positive
- The distance the amino acid species moves from the starting point depends on the size of the ion
- Larger ions move more slowly
 - A** has moved the furthest, so is the smallest species
 - B** has moved the least, so is the largest species
- You can use numbers to identify the relative sizes

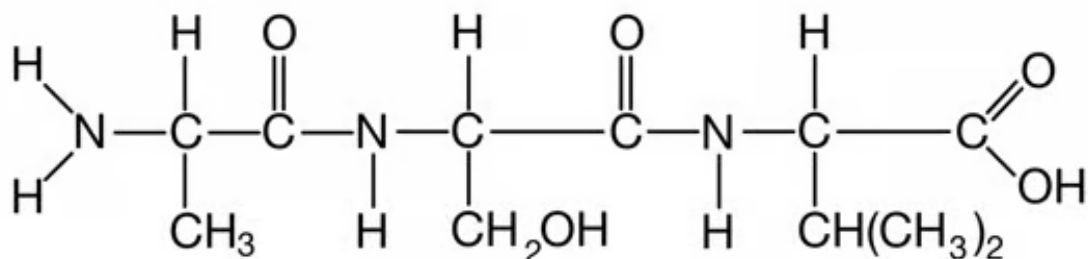
5d

d)

i) The number of different tripeptides that can be made using **one** molecule of **each** of the amino acids in Table 5.1 is:

- 6 / six; [1 mark]

ii) The structure of the tripeptide, showing the peptide bonds in displayed form, is:



- Two displayed peptide bonds; [1 mark]
- Correct formula of peptide; [1 mark]

[Total: 3 marks]

- The three amino acids can be arranged in six different ways:
 - alanine-serine-valine
 - alanine-valine-serine
 - serine-alanine-valine
 - serin-valine-alanine
 - valine-alanine-serine
 - valine-serine-alanine
- You would get credit for any one of these tripeptides, so the central R groups of each amino acid can be shown in any order
- You can write this as a structural formula but with the peptide bond, -NHCO- shown in displayed form

